Closing the logic gap – expanding who we consider for TB treatment

Anne Williamson[1], Lara D Veeken[2], Reinout van Crevel[2], Chris Cousins[3], Hanif Esmail[4], Katherine C Horton[5], Raspati C Koesoemadinata[6], Kate LeGrand[5], Matthew Saunders[3], Ingrid Schoeman[7], Sean Wasserman[1], Rein MGJ Houben[5]*

[1]Queen Mary and Barts Health TB Centre, Queen Mary University of London, London
[2]UK Department of Internal Medicine and Radboud Center for Infectious Diseases, Radboud University Medical Center, Nijmegen
[3]Institute for Infection and Immunity, City St George's, University of London, London, United Kingdom
[4]WHO Collaborating Centre for Tuberculosis Research and Innovation, UCL Centre for Global Tuberculosis Research, Institute for Global Health, University College London, London, UK
[5]TB Modelling Group, TB Centre, London School of Hygiene and Tropical Medicine, London, UK
[6]Research Center for Care and Control of Infectious Disease, Universitas Padjadjaran, Bandung, Indonesia;
[7]TB Proof, Cape Town, South Africa
**Corresponding Author, Rein.Houben@lshtm.ac.uk*

Appendix Table: detailed definitions and sources

	Microbiologically confirmed TB			% offered treatment at baseline evaluation		
	%	**Calculations**	**Source**	**%**	**Calculations**	**Source**
Household contact (HHC)	3-4%	HHC defined in the source as individuals sharing living space or prolonged exposure to an index case during the preceding 3 months. 3.2% pooled prevalence of microbiologically confirmed TB across all contacts. Among HHC specifically, 3.5% pooled TB prevalence (not restricted to microbiologically confirmed disease).	Systematic review of 106 studies with up to five years follow-up (n= 955,733)[1]	55%	Value from treatment initiation rate amongst eligible HHC of 101,325/185,502, where eligible HHC were defined according to national guidelines including those with normal X-ray and positive TST or IGRA, or 'free of active disease' with TBI testing unavailable. Treatment initiation taken as a conservative lower bound estimate of % offered (noting a higher proportion of individuals may have been offered but declined treatment).	Large Indian state-level evaluation (n= 406,291)[2]
Trace PCR+/ culture -*	27%	Of 128 trace-positive participants, 27 were culture-positive. 9/101 (8.9%) had non-culture microbiological confirmation other than the trace PCR result at baseline. Among those not diagnosed at baseline and followed up, 15/75 (20%) developed microbiologically-confirmed TB (including culture, urine LAM, or second positive Ultra) during 2-year follow-up. Estimated total: 8.9%+(1−8.9%)×20% = 27%.	Analysis of 128 trace-positive participants through Ugandan community-based screening study[3]	18%	18/101 (17.8%) trace-positive, culture-negative participants offered treatment at baseline (derived from 45 recommended treatment at baseline, of whom 27 were culture-positive).	Analysis of 128 trace-positive participants through Ugandan community-based screening study[3]

CXR+/ culture -	19-58%	Untreated individuals with TB-suggestive x-ray and negative culture. Range derived from pooled studies reporting progression to culture-positive TB over follow-up (mostly 5 years).	Systematic review of 13 studies (n=32,568)[4]	6%	28/456 (6.1%) bacteriologically negative individuals with abnormal chest x-ray initiated TB treatment. Treatment initiation taken as a conservative lower bound estimate of % offered (noting a higher proportion of individuals may have been offered but declined treatment).	Systematic review and individual patient data meta-analysis (n=15,121)[5]

*For the purposes of this analysis, trace PCR+/culture- results are not counted as 'microbiologically-confirmed TB', as they are presented as a distinct phenotype; however, WHO classifies Xpert Ultra trace-positive results as microbiological confirmation.

Appendix references:

[1] Velen K, Shingde RV, Ho J, Fox GJ. The effectiveness of contact investigation among contacts of tuberculosis patients: a systematic review and meta-analysis. *Eur Respir J.* 2021 Dec;58(6):2100266.

[2] Mundra A, Bhatnagar T, Das M, Sangale S, Patil H, Raut A, Kadu A, Paradkar RJ, Gupta SS, Garg BS. Identifying missed opportunities in tuberculosis preventive treatment care cascade: Analysis of programme data from Maharashtra, India. PLOS Glob Public Health. 2026 Jan 16;6(1):e0004630.

[3] Sung J, Nantale M, Nalutaaya A, Biché P, Mukiibi J, Akampurira J, et al. Long-term risk of tuberculosis among individuals with Xpert Ultra trace screening results in Uganda: a longitudinal follow-up study. *The Lancet Infectious Diseases.* 2026 Feb 1;26(2):203–12.

[4] Gray AT, Macpherson L, Carlin F, Sossen B, Richards AS, Kik SV, et al. Treatment for radiographically active, sputum culture-negative pulmonary tuberculosis: A systematic review and meta-analysis. *PLOS ONE.* 2023 Nov 16;18(11):e0293535.

[5] Kim S, Can MH, Agizew TB, Auld AF, Balcells ME, Menzies NA, et al. Factors associated with tuberculosis treatment initiation among bacteriologically negative individuals evaluated for tuberculosis: An individual patient data meta-analysis. PLoS Med. 2025 Jan 13;22(1):e1004502.